WORKED SOLUTION

STEP I 2014, Question 10

Mechanics • Collisions and Restitution

STEP I 2014, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q10

The key idea

Both parts are energy for the free flight and Newton's experimental law for each impact. In part (ii) the trick is to notice that at the moment of the ball-on-ball collision the big ball has just reversed direction, while the small ball is still coming down at full speed: the two meet with a large closing speed, and a heavy lower ball can hurl the light one far higher than it started. After that it is a two-equation collision problem, and the given numbers make the ratio of speeds come out cleanly.

Solution

Model both balls as uniform rigid spheres moving vertically, with no air resistance, and take g as the acceleration due to gravity. Each impact is instantaneous, so gravity does not change any momentum during it.

Part (i)

The ball touches the ground when its centre is at height R, so it has fallen a distance H. By conservation of energy its speed just before impact is

$$u = \sqrt{2gH}.$$

The ground is fixed, so Newton's experimental law gives a rebound speed of eu upwards. The ball then rises until its speed is zero; by energy again it rises a further

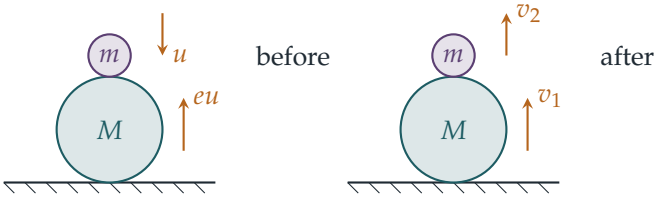
$$\frac{(eu)^2}{2g} = \frac{e^2 \cdot 2gH}{2g} = e^2 H.$$

Its centre starts this rise at height R, so it reaches a height $R + He^2$ above the ground, as required.

Part (ii)

Both balls fall freely from rest together, so at every instant they have the same speed. The lower ball reaches the ground after its centre has fallen H, so at that moment both balls are moving downwards at $u = \sqrt{2gH}$. The lower ball bounces and, as in part (i), leaves the ground at speed eu upwards; immediately afterwards it meets the upper ball, which is still moving down at u.

Take upwards as positive, and let the velocities just after the collision between the balls be v_1 (ball of mass M) and v_2 (ball of mass m).



Conservation of momentum gives

$$Meu - mu = Mv_1 + mv_2,$$

and Newton's experimental law (speed of separation is e times speed of approach, where the approach speed is $eu + u$) gives

$$v_2 - v_1 = e(1+e)u.$$

Substituting $v_1 = v_2 - e(1+e)u$ into the momentum equation,

$$\begin{aligned}(M+m)v_2 &= Meu - mu + Me(1+e)u \\ &= (Me(2+e) - m)u,\end{aligned}$$

so

$$v_2 = \frac{Me(2+e) - m}{M+m}\,u.$$

? **Ask yourself** Where is the centre of the small ball when the collision happens?

The collision is immediately after the bounce, so the big ball is still touching the ground: its centre is at height R, its top at $2R$, and the small ball's centre at $2R + r$. From there the small ball rises a further $v_2^2/(2g)$, so

$$h = 2R + r + \frac{v_2^2}{2g}.$$

With the given values, $2R + r = 0.45$, so $v_2^2/(2g) = 4.05$; and $u^2/(2g) = H = 1.8$. Dividing,

$$\frac{v_2^2}{u^2} = \frac{4.05}{1.8} = \frac{9}{4}, \qquad \text{so} \qquad v_2 = \tfrac{3}{2}u,$$

taking the positive root because the small ball moves upwards after the collision.

With $e = \frac{2}{3}$ we have $e(2+e) = \frac{2}{3} \cdot \frac{8}{3} = \frac{16}{9}$. Writing $k = M/m$ and dividing top and bottom by m,

$$\frac{\frac{16}{9}k - 1}{k+1} = \frac{3}{2} \implies \tfrac{32}{9}k - 2 = 3k + 3 \implies \tfrac{5}{9}k = 5,$$

so

$$\frac{M}{m} = 9.$$

Checks

- With $M/m = 9$ we get $v_2 = \frac{3}{2}u$ and $v_1 = v_2 - \frac{10}{9}u = \frac{7}{18}u$. Momentum before is $9m \cdot \frac{2}{3}u - mu = 5mu$, and after is $9m \cdot \frac{7}{18}u + m \cdot \frac{3}{2}u = 5mu$. Both agree.
- Since $v_1 < v_2$ the balls separate after the impact, as they must.
- Numerically, $u = \sqrt{2 \times 9.8 \times 1.8} \approx 5.94\,\mathrm{m\,s^{-1}}$ and $v_2 \approx 8.91\,\mathrm{m\,s^{-1}}$, which rises 4.05 m, giving $h = 4.5$.
- If $M \gg m$, $v_2 \to e(2+e)u$; with $e = 1$ this is $3u$, the well-known factor of nine in height for a perfectly elastic "stacked ball" drop.

Common slips

- Measuring the final height from the ground as $v_2^2/(2g)$ alone, forgetting that the collision happens with the small ball's centre at $2R + r$.
- Using $e \cdot 2u$, or eu, as the separation speed: the approach speed is $eu + u$, because the balls are moving towards each other.
- Sign errors: fix upwards as positive and give the upper ball velocity $-u$ before the collision.

Beyond STEP

Stacked-ball drops are a lovely demonstration of how momentum can be funnelled into a light object: with a tower of several balls, each lighter than the one beneath, the top ball can be launched to many times its release height. The cleanest way to analyse such collisions is in the zero-momentum (centre-of-mass) frame, where each ball's velocity is simply reversed and scaled by e. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops collisions and the centre-of-mass frame carefully.

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